

Chapter 4

Queuing System

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Queuing System

Introduction

- A queueing system consists of one or more servers that provide service of some kind to arriving customers.
- If the servers are busy, the customers join one or more queues in front of the servers, hence forming a queueing system.

System	Servers	Customers
Bank	Tellers	Customers
Hospital	Doctors, Nurses, beds	Patients
Computer System	CPU, I/O devices	Jobs
Manufacturing systems	Machines, Workers	Parts

Queuing System

- Waiting line queues are one of the most important areas, where the technique of simulation has been extensively employed.
- The waiting lines or queues are a common site in real life.
- People at railway ticket window, vehicles at a petrol pump or at a traffic signal, products at a machining center, television sets at a repair shop are a few examples of waiting lines.

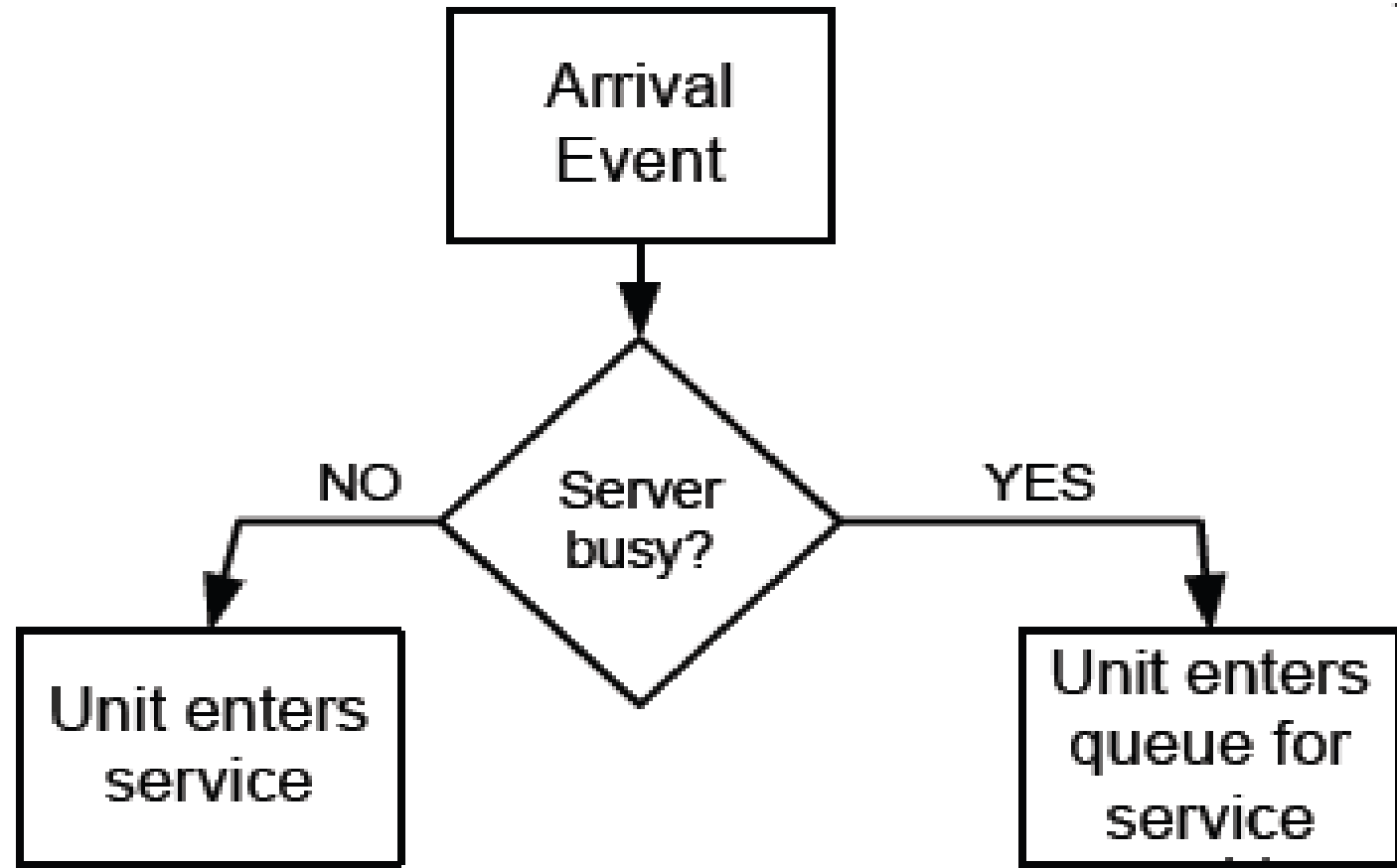
Queuing System

- The waiting line situations arise, either because,
 - There is too much demand on the service facility so that the customers or entities have to wait for getting service, or
 - There is too less demand, in which case the service facility have to wait for the entities.
- The objective in the analysis of queuing situations is to balance the waiting time and idle time, so as to keep the total cost at minimum.

Queuing System

- The queuing theory gives the credit of its development to an engineer A.K. Erlang, who in 1920, studied waiting line queues of telephone calls in Copenhagen, Denmark.
- The problem was that during the busy period, telephone operators were unable to handle the calls, there was too much waiting time, which resulted in customer dissatisfaction.

Queuing System



4.1 Elements of Queuing System

Elements of Queuing Systems

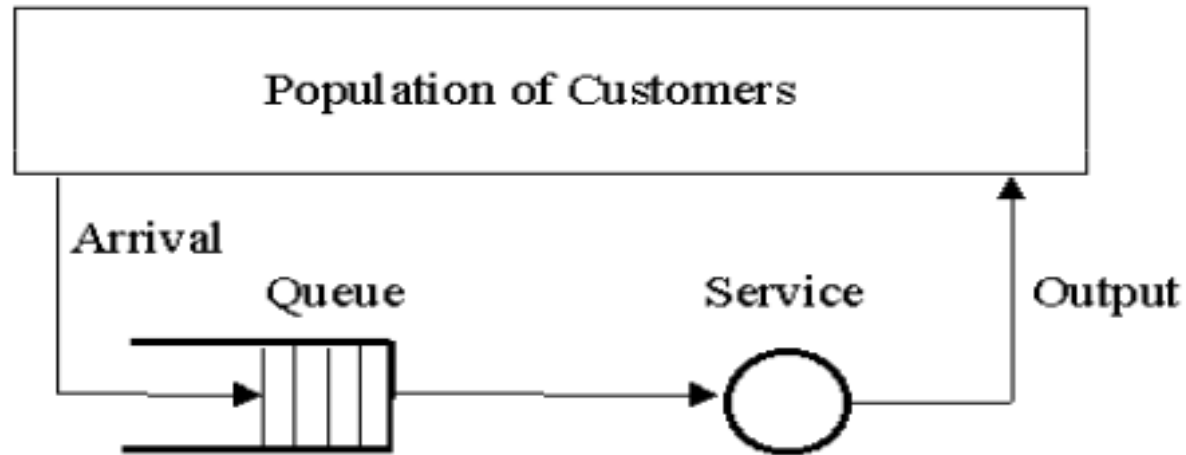


Figure 1

Elements of Queuing System

1. Population of Customers or calling source

- can be considered either limited (closed systems) or unlimited (open systems).
- Unlimited population represents a theoretical model of systems with a large number of possible customers (a bank on a busy street, a motorway petrol station).
- Example of a limited population may be a number of processes to be run (served) by a computer or a certain number of machines to be repaired by a service man.
- It is necessary to take the term "customer" very generally.
- Customers may be people, machines of various nature, computer processes, telephone calls, etc

4.1 Elements of Queuing System

2. Arrival

- defines the way customers enter the system.
- Mostly the arrivals are random with random intervals between two adjacent arrivals.
- Typically the arrival is described by a random distribution of intervals also called Arrival Pattern.

4.1 Elements of Queuing System

3. Queue or waiting line

- represents a certain number of customers waiting for service (of course the queue may be empty).
- Typically the customer being served is considered not to be in the queue.
- Sometimes the customers form a queue literally (people waiting in a line for a bank teller).
- Sometimes the queue is an abstraction (planes waiting for a runway to land).
- There are two important properties of a queue: Maximum Size and Queuing Discipline.

4.1 Elements of Queuing System

- **Maximum Queue Size** (also called System capacity) is the maximum number of customers that may wait in the queue (plus the one(s) being served).
- Queue is always limited, but some theoretical models assume an unlimited queue length.
- If the queue length is limited, some customers are forced to renounce without being served.
- **Queuing Discipline** includes FIFO, LIFO, SPT (Shortest Processing Time First), etc.

4.1 Elements of Queuing System

4. Service

- represents activity which takes time and the customers are waiting for.
- It may be a real service carried on persons or machines, but it may be a CPU time slice, connection created for a telephone call, being shot down for an enemy plane, etc.

5. Output

- represents the way customer leaves the system
- Output is mostly ignored by theoretical models, but sometimes the customers leaving the server enter the queue again ("round robin" time-sharing systems).

4.2 Characteristics of Queuing System

1. The Calling Population

- The population of potential customers in the system.
- May be assumed to be finite or limited (closed system) and infinite or unlimited (open system).
- Open system represents model of a system with large number of possible customers such as in banking system, restaurant system, etc.
- Here, the arrival rate to the queuing system is not affected by the number of customers who have left the calling population and joined the queuing system.
- Finite population may be a number of process to be run by a computer with a fixed number of processor. The arrival rate to the queuing system depends on the number of customers being served and waiting.

4.2 Characteristics of Queuing System

2. Arrival Pattern

- It defines the way customer enters into the queuing system.
- Mostly arrivals are random with random intervals between two adjacent arrival.
- Customers may arrive one at a time or in batches. The batches may be of constant size or of random size.
- In an infinite population the arrival rate is not affected by the number of customers who have left the calling population.
- When the arrival process is homogeneous over time the arrival rate is usually assumed to be constant.
- One of the model of random arrivals is the Poisson arrival process which has been successfully employed as a model of the arrival of people at restaurant, drive-in-banks, and other service facilities ; the arrival of telephone calls to a call center, the arrival of demand or order of a product, etc.

4.2 Characteristics of Queuing System

2. Arrival Pattern

- A second important class of arrivals is scheduled arrivals such as patients to a physician's or scheduled airline flight arrivals to an airport.
- A third situation occurs when at least one customer is assumed to always be present in the queue so that the server is never idle because of lack of customers. For example, the customers may represent raw material for a product, and sufficient raw material is always assumed to be available.

4.2 Characteristics of Queuing System

3. System capacity

- In many queuing systems, there is a limit to the number of customers that may be in the queue.
- Queue represents a certain number of customers waiting for service.
- Maximum number of customers that may be wait in the queue is the system capacity or the maximum size of queue.
- For simplicity, queue is always made limited in most of the system.

4.2 Characteristics of Queuing System

4. Queue behavior and discipline

- Queue behavior refers to the actions of customers while in a queue waiting for a service to begin.
- In some situations, there is a possibility that incoming customers will balk (leave when they see that the line is too long), renege (leave after being in the line when they see that the line is moving too slowly), or jockey (move from one line to another if they think they have chosen a slow line).
- Queuing Discipline refers to the logical ordering of customers in a queue and determines which customer will be chosen for service when a server becomes free. There are these ways:
 - 1) FIFO (First In First Out) also called FCFS (First Come First Serve) - orderly queue.
 - 2) LIFO (Last In First Out) also called LCFS (Last Come First Serve) - stack.
 - 3) SIRO (Serve In Random Order).
 - 4) Shortest Processing Time First (SPT)
 - 5) Service according to Priority (PR)

4.2 Characteristics of Queuing System

5. Service

- represents some activity that takes time and that the customers are waiting for. Again, take it very generally.
- It may be a real service carried on persons or machines, but it may be a CPU time slice, connection created for a telephone call, being shot down for an enemy plane, etc.
- Typically a service takes random time.

4.2 Characteristics of Queuing System

6. Output

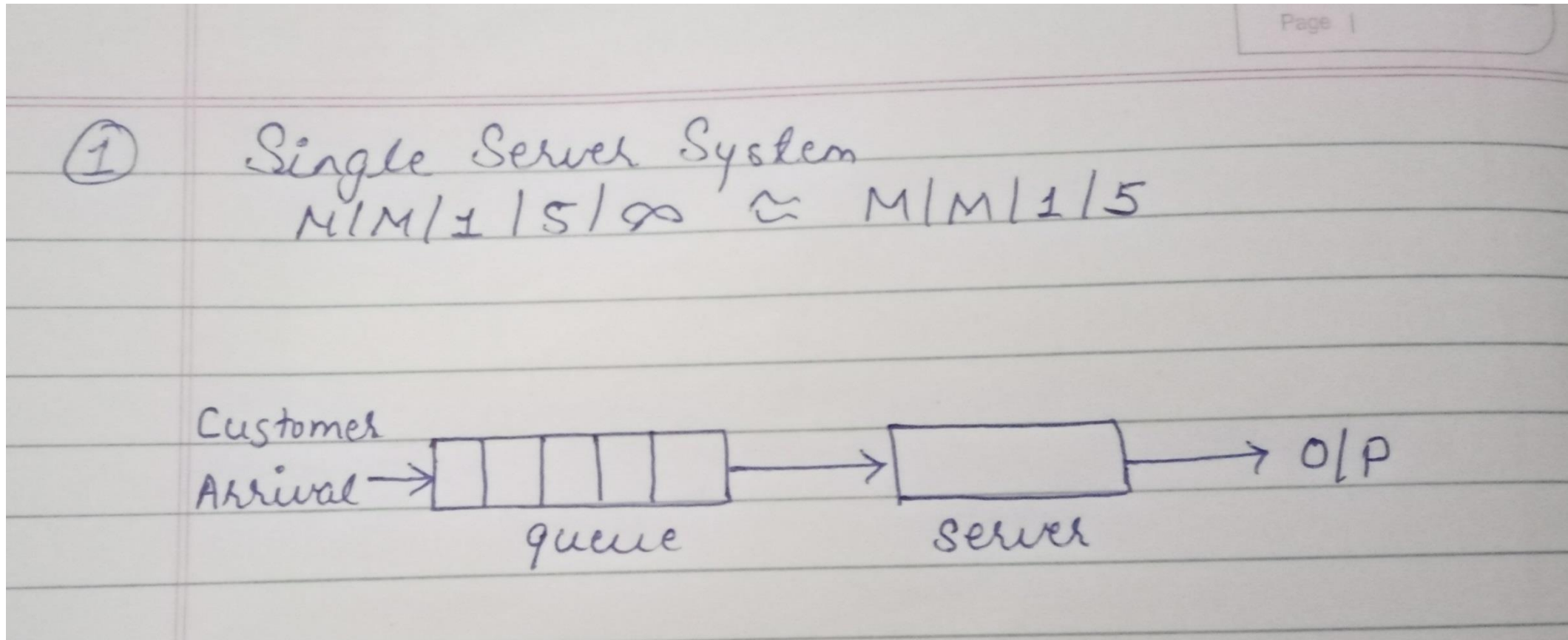
- Output represents the way customers leave the system.
- Output is mostly ignored by theoretical models, but sometimes the customers leaving the server enter the queue again ("round robin" time-sharing systems).

4.3 Model of Queuing System

- Model of a queue can be categorized as:
 1. Single Server Queue System
 2. Multiple Server Queue System
 - Single Queue Multiple Server System
 - Multiple Queue Multiple Server System

4.3 Model of Queuing System

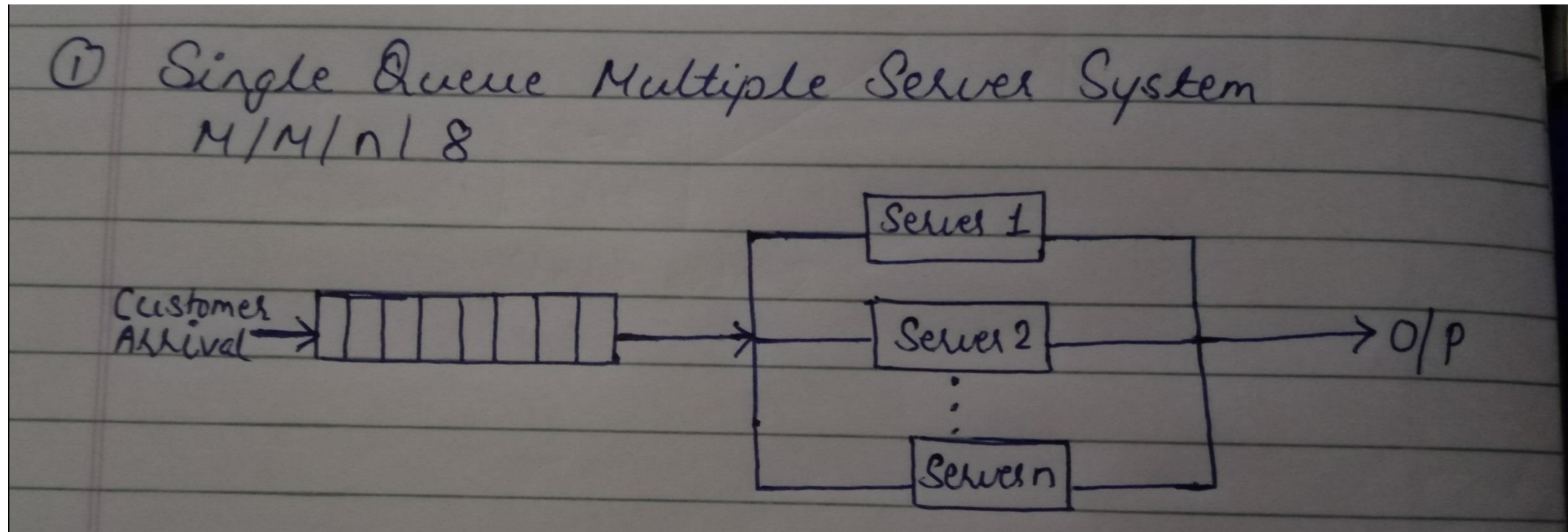
1. Single Server Queue System



4.3 Model of Queuing System

2. Multiple Server Queue System

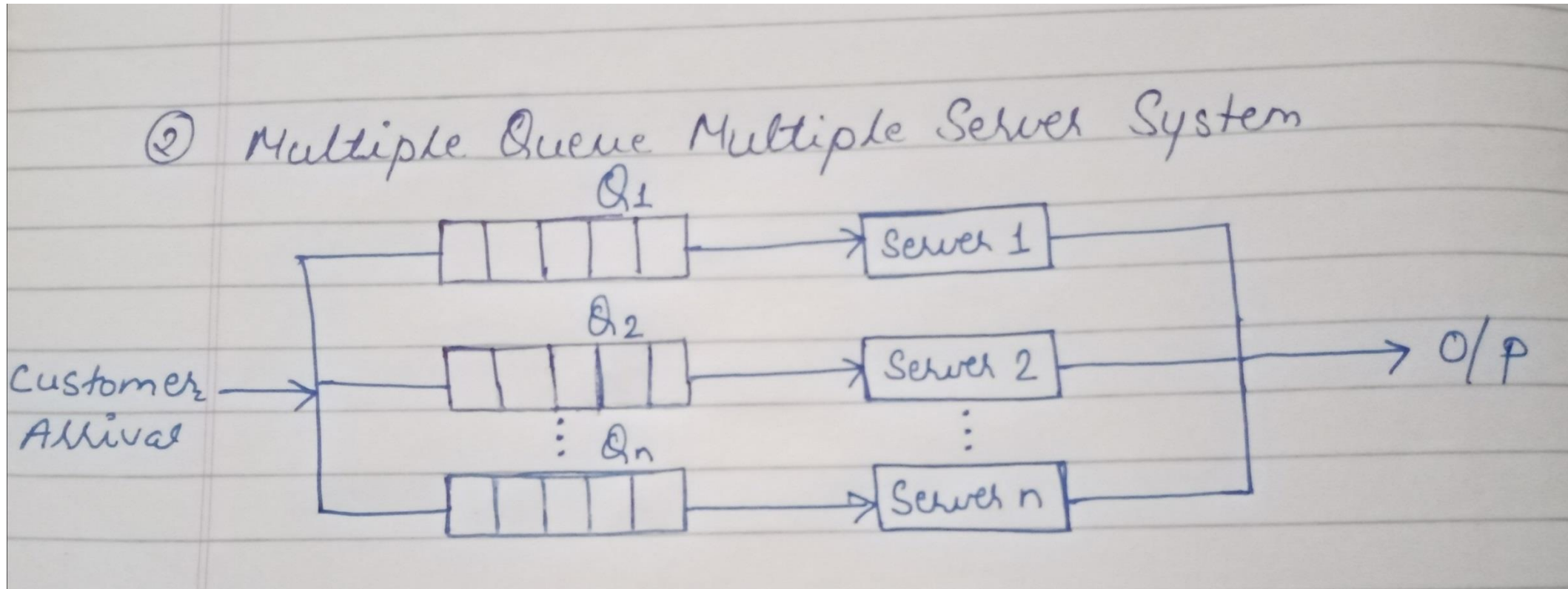
a. Single Queue Multiple Server System



4.3 Model of Queuing System

2. Multiple Server Queue System

b. Multiple Queue Multiple Server System



4.4 Types of Queuing System

1. Structured Queue

- Here, queue is formed in a fixed, predictable position, such as at supermarket checkouts, and other retail locations such as banks or airport security.

2. Unstructured Queue

- Here, queue is formed in unpredictable and varying locations and directions. This is often in the case of some forms of retail, taxi queues, ATMs and at periods of high demand in many situations.

3. Kiosk Based Queue

- Often used for medical, banking and many governmental services locations. As people arrive, they enter basic information into a kiosk about themselves and reason for their visit. The information is organized and presented to staff to allow for faster customer service response.

4.5 Queueing Notations

- Kendall has proposed a notational system for queue in 1953. Its several modifications is available however the most comprehensive classification uses 6 symbols: $A/B/s/q/c/p$ where:
 - A is the arrival pattern (distribution of intervals between arrivals).
 - B is the service pattern (distribution of service duration).
 - s is the number of servers.
 - q is the queuing discipline (FIFO, LIFO, ...). Omitted for FIFO or if not specified.
 - c is the system capacity. Omitted for unlimited queues.
 - p is the population size (number of possible customers). Omitted for open systems.

Kendall Classification of Queuing Systems

- These symbols are used for arrival and service patterns:
 - M is the Poisson (Markovian) process with exponential distribution of intervals or service duration respectively.
 - E_m is the Erlang distribution of intervals or service duration.
 - D is the symbol for deterministic (known) arrivals and constant service duration.
 - G is a general (any) distribution

Kendall Classification of Queuing Systems

Examples:

1. D/M/1

- Deterministic (known) input,
- one exponential server,
- one unlimited FIFO or unspecified queue,
- unlimited customer population.

Kendall Classification of Queuing Systems

2. M/G/3/20

- Poisson input,
- three servers with any distribution,
- maximum number of customers that can be served is 20,
- unlimited customer population.

Kendall Classification of Queuing Systems

3. D/M/1/LIFO/10/50

- Deterministic arrivals,
- one exponential server,
- queue is a stack of the maximum size 9,
- total number of customers 50.

Kendall Classification of Queuing Systems

4. M/M/3/FCFS/20/1500

- Time between successive arrivals is exponentially distributed
- Service times are exponentially distributed
- Three servers
- Service discipline is first-come-first -served
- Maximum customers to be served is 20 i.e. 20 buffers = 3 service + 17 waiting.
- Total of 1500 jobs are waiting to be serviced

Kendall Classification of Queuing Systems

5. M/D/6/20/1000/LIFO (2081 Chaitra)

6. D/M/6/FIFO/25/3000 (2077 Chaitra)

4.6 Measures of system performance

- The performance of a queuing system can be evaluated in terms of a number of response parameters, however the following four are generally employed.
 1. Average number of customers in the queue or in the system
 2. Average waiting time of the customers in the queue or in the system
 3. System utilization
 4. The cost of the waiting time & idle time

4.6 Measures of System Performance

- The knowledge of average number of customers in the queue or in the system helps to determine the space requirements of the waiting entities. Also too long a waiting line may discourage the prospectus customers, while no queue may suggest that service offered is not of good quality to attract customers.
- The knowledge of average waiting time in the queue is necessary for determining the cost of waiting in the queue.

System/Server Utilization

- System Utilization is the percentage of the system capacity that has been utilized which reflects that extent to which the facility is busy rather than idle.
- System utilization factor (s) is the ratio of average arrival rate (λ) to the average service rate (μ).
 $S = \lambda/\mu$ in the case of a single server model
 $S = \lambda/\mu n$ in the case of a “n” server model
- The system utilization can be increased by increasing the arrival rate which amounts to increasing the average queue length as well as the average waiting time. Under the normal circumstances 100% system utilization is not a realistic goal.

M/M/1 Queue

- The most commonly used type of queue.
- Used to model single processor systems or individual devices in a computer system.

Assumption

- Interarrival rate of λ -> exponentially distributed
- Service rate of μ -> exponentially distributed
- Single server
- FCFS
- Unlimited queue lengths allowed
- Infinite number of customers

Analysis of M/M/1 queue

- Given:

λ : Arrival rate of jobs (packets on input link)

μ : Service rate of the server (output link)

System utilization factor (s) = λ / μ

Solve:

L : average number of customers in queuing system: $L = \lambda / (\mu - \lambda)$

L_q : average number in the queue: $L_q = \lambda^2 / (\mu(\mu - \lambda))$

W : average waiting time in whole system: $W = L / \lambda$

W_q : average waiting time in the queue: $W_q = W - (1 / \mu)$

Analysis of M/M/1 queue: Example 1

- Example 1: If arrivals are occurring at rate $\lambda = 10$ per hour and management has the choice of two servers, one who works at rate $\mu_1 = 11$ customers per hour and the second at rate $\mu_2 = 12$ customers per hour, then:

➤ The respective system utilization would be: $s_1 = \lambda / \mu_1 = 10/11 = 0.909$ and $s_2 = \lambda / \mu_2 = 10/12 = 0.833$.

➤ With the first server the average number in the system would be:

$$L_1 = \lambda / (\mu - \lambda) = 10$$

And with the second server the average number in the system would be:

$$L_2 = \lambda / (\mu - \lambda) = 5$$

Analysis of M/M/1 queue: Example 1

➤ With the first server the average number in the queue would be:

$$L_q = \lambda^2 / (\mu(\mu - \lambda)) = 100 / (11(11 - 10)) = 100 / 11 = 9.09$$

And with the second server the average number in the queue would be:

$$L_q = \lambda^2 / (\mu(\mu - \lambda)) = 100 / (12(12 - 10)) = 100 / 24 = 4.16$$

Analysis of M/M/1 queue: Example 1

➤ With the first server the average waiting time in the system would be:

$$W_1 = L_1 / \lambda = 10/10 = 1$$

And with the second server the average waiting time in the system would be:

$$W_2 = L_2 / \lambda = 5/10 = 0.5$$

Analysis of M/M/1 queue: Example 1

➤ With the first server the average waiting time in the queue would be:

$$\begin{aligned}W_q &= W - (1/\mu_1) \\ &= 1 - (1/11) = 0.90\end{aligned}$$

And with the second server the average waiting time in the queue would be:

$$\begin{aligned}W_q &= W - (1/\mu_2) \\ &= 0.5 - (1/12) = 0.4167\end{aligned}$$

➤ Since, the number of customer in the system is less in case of second server and also the waiting time is less, there is more prospect of new customers visiting the system. Also, the server utilization factor between both the server is not significantly different, choosing second server would be a good decision.

Analysis of M/M/1 queue: Example 2

- Example 2: For the M/M/1 queue with service rate $\mu = 10$ customers per hour, consider how L and W increase as the arrival rate, λ increases from 5 to 8.64 by increments of 20% and then to $\lambda = 10$.

Analysis of M/M/1 queue: Example 2

- Example 2: For the M/M/1 queue with service rate $\mu = 10$ customers per hour, consider how L and W increase as the arrival rate, λ increases from 5 to 8.64 by increments of 20% and then to $\lambda = 10$.

λ	5.0	6.0	7.2	8.64	10.0
s	0.500	0.600	0.720	0.864	1.0
L	1.00	1.50	2.57	6.35	Infinity
w	0.20	0.25	0.36	0.73	infinity

Analysis of M/M/1 queue:

Customers arrive at a bank according to a Poisson process with a **mean interarrival time of 10 minutes**. Customers spend an **average of 5 minutes** receiving service at a **single available counter**.

Assuming an **M/M/1 queueing system**, answer the following:

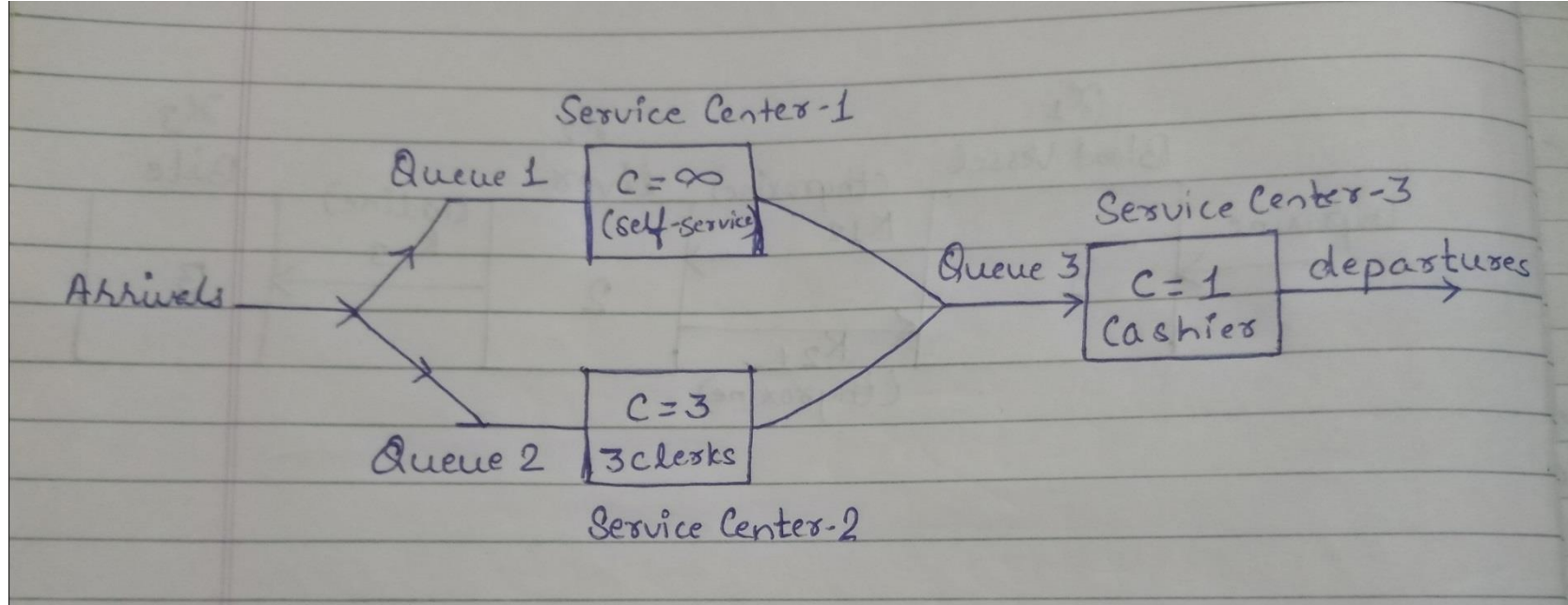
1. What is the probability that an arriving customer **will not have to wait** at the counter?
2. What is the **expected number of customers in the bank** (including any customer being served)?
3. How much time can a customer **expect to spend in the bank**, including both waiting time and service time?

Analysis of M/M/1 queue:

A barber in a one-man shop takes exactly 30 minutes to complete a haircut. Customers arrive according to a Poisson process at an average rate of one customer every 40 minutes. How long, on average, must a customer wait before receiving service? (2079 Chaitra)

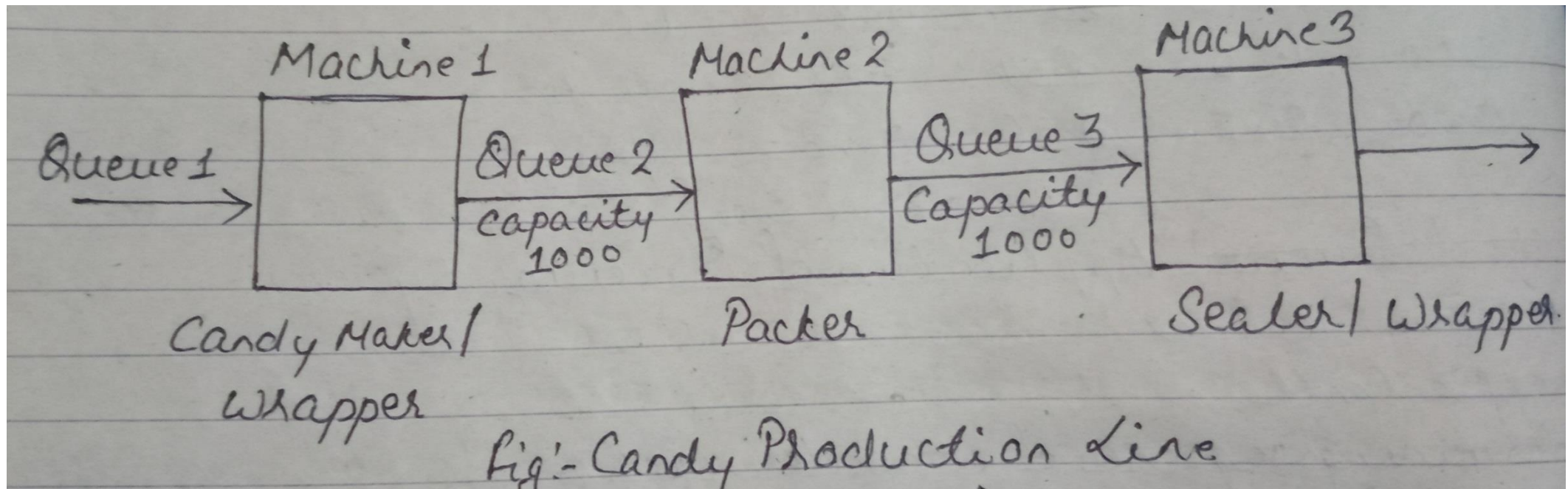
4.7 Networks of Queue

- In real world implementation, many system are naturally modeled as networks of single queues in which customers departing from one queue may be routed to another.
- Example 1: Consider a discount warehouse where customers may either serve themselves or wait for one of three clerks, then finally leave after paying a single cashier.



4.7 Networks of Queue

- Example 2: A candy manufacturer has a production line that consists of 3 machines separated by inventory-in-process buffers. The first machine makes and wraps the individual pieces of candy, the 2nd packs 50 pieces in a box, and 3rd machine seals and wraps the box. The two inventory buffers have capacities of 1000 boxes each.



4.7 Networks of Queue

- The system is modeled as having 3 service centers, each center having $c=1$ server, with queue capacity constraints between machines.
- It is assumed that a sufficient supply of raw material is always available at the first queue.
- Because of the queue capacity constraints, machine 1 shuts down whenever its inventory buffer (queue 2) fills to capacity, and machine 2 shuts down whenever its buffer empties.
- In brief, the system consists of 3 single-server queues in series queue capacity constraints and a continuous arrival stream at the first queue.

4.8 Application of Queuing System

- Telecommunications
- Traffic control
- Determining the sequence of computer operations
- Health services (e.g. control of hospital bed assignments)
- Airport traffic, airline ticket sales
- Layout of manufacturing systems.

References

- Jerry Banks, John S. Carson, II Barry L. Nelson , David M. Nicol, P. Shahabudeen “Discrete Event system simulation”